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QUESTION 1: RAIN AND WET GROUND

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Goal: Ground is wet

Initial Clauses:

C1: R

C2: $W \vee \sim R$

C3: $\sim W$

Resolve (R) and $(W \vee \sim R) \rightarrow W$

Resolve $(W \vee \sim R)$ and $(\sim W) \rightarrow \sim R$

Resolve (R) and $(W \vee \sim R) \rightarrow W$

Resolve (R) and $(\sim R) \rightarrow \square$

Result: Empty Clause ($\square$) obtained.

Therefore, the goal is PROVED.

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QUESTION 2: STUDENT ASSIGNMENT SUBMISSION

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Goal: Rahul receives internal marks

Initial Clauses:

C1: $\sim M$

C2: S

C3: $M \vee \sim S$

Resolve $(\sim M)$ and $(M \vee \sim S) \rightarrow \sim S$

Resolve (S) and $(M \vee \sim S) \rightarrow M$

Resolve $(\sim M)$ and (M) $\rightarrow \square$

Result: Empty Clause ($\square$) obtained.

Therefore, the goal is PROVED.

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QUESTION 3: LIBRARY MEMBERSHIP

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Goal: Priya can borrow books

Initial Clauses:

C1: $\sim B$

C2: $B \vee \sim L$

C3: L

Resolve $(\sim B)$ and $(B \vee \sim L) \rightarrow \sim L$

Resolve $(B \vee \sim L)$ and (L) $\rightarrow B$

Resolve $(B \vee \sim L)$ and (L) $\rightarrow B$

Resolve ($\sim L$) and (L) $\rightarrow \square$
Result: Empty Clause ($\square$) obtained.
Therefore, the goal is PROVED.

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QUESTION 4: PLACEMENT ELIGIBILITY

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Goal: Arun is eligible for placement

Initial Clauses:

C1: A

C2: $P \vee \sim A$

C3: $\sim P$

Resolve (A) and ($P \vee \sim A$) $\rightarrow P$

Resolve ($P \vee \sim A$) and ($\sim P$) $\rightarrow \sim A$

Resolve ($\sim A$) and (A) $\rightarrow \square$

Result: Empty Clause ($\square$) obtained.

Therefore, the goal is PROVED.

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QUESTION 5: ACCESS CONTROL SYSTEM

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Goal: User is granted access

Initial Clauses:

C1: C

C2: $\sim G$

C3: $G \vee \sim A$

C4: $A \vee \sim C$

Resolve (C) and ($A \vee \sim C$) $\rightarrow A$

Resolve ($\sim G$) and ($G \vee \sim A$) $\rightarrow \sim A$

Resolve ($G \vee \sim A$) and ($A \vee \sim C$) $\rightarrow G \vee \sim C$

Resolve ($\sim A$) and ($A \vee \sim C$) $\rightarrow \sim C$

Resolve ($\sim A$) and (A) $\rightarrow \square$

Result: Empty Clause ($\square$) obtained.

Therefore, the goal is PROVED.

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FINAL SUMMARY

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Q1 - Ground is wet	: PROVED
Q2 - Rahul receives internal marks	: PROVED
Q3 - Priya can borrow books	: PROVED
Q4 - Arun is eligible for placement	: PROVED
Q5 - User is granted access	: PROVED

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